

What are you trying to do?

You can dip into this guide as needed. Find your task below and tap its page number.

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My equipment	3	Reuse settings for a drone or test rig
Check a connection	4	Check incoming data over USB or UDP
Check position	6	Check whether a position is fresh
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Follow the numbers

Each screenshot has numbered arrows matching the steps beside it. Use the quick-start guide for your first session; use this guide when you need a particular tool. Screenshots use example equipment and practice data.

Find a tool and keep it handy

You can start from Home or search directly. Use whichever is quicker for the job.

1 Search in ordinary words

Tap Tools or Search. GPS finds tools for coordinates and position; battery finds the runtime calculator.

2 Open the result

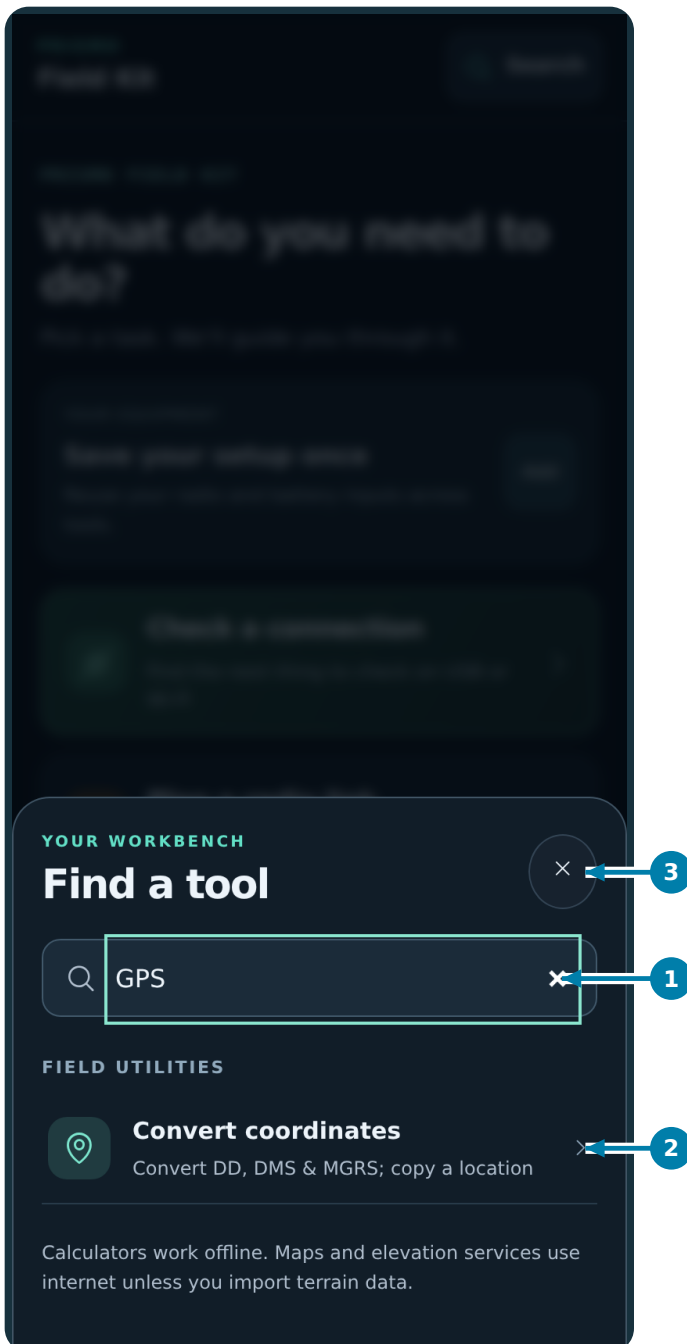
Tap a matching tool. If you use it often, tap the star in its header to pin it on Home.

3 Close the list to return

Close the search list to get back to your screen. Android Back also closes lists, then returns an open tool to Home.

Four places to remember

Home starts a task. Equipment holds your saved setup. Saved checks holds snapshots. Tools gives you the full list.

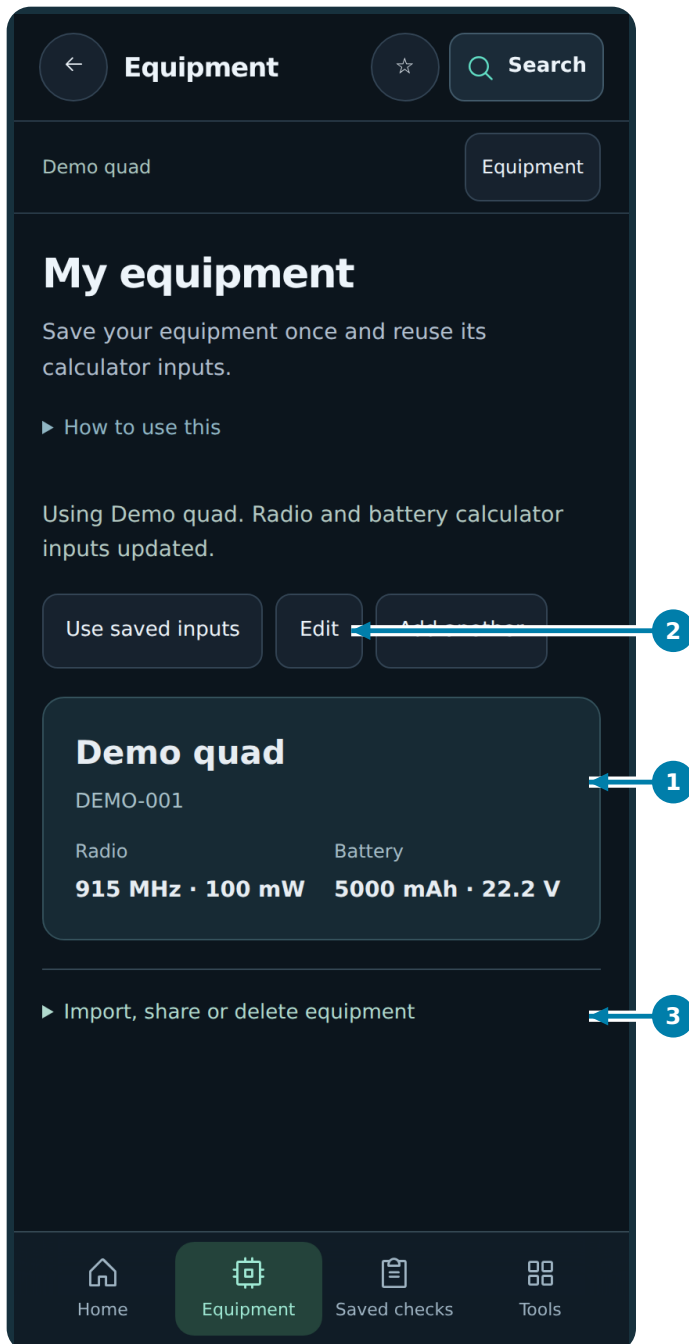


On a small screen

Swipe up to reach more controls. Tap section headings to expand them. The bottom navigation may hide while the keyboard is open.

Keep your equipment settings together

Use this when you work on more than one drone, or want to stop re-entering the same numbers.



1 Check which setup is selected

The summary shows the name, serial number and main radio and battery values.

2 Edit what changed

Open Edit, change the relevant fields, then save. The radio, battery and hardware fields are grouped to make them easier to find.

3 Move a setup to another device

Open Import, share or delete equipment. Copy equipment JSON exports the setup; Import reads that format back in.

When the numbers look different

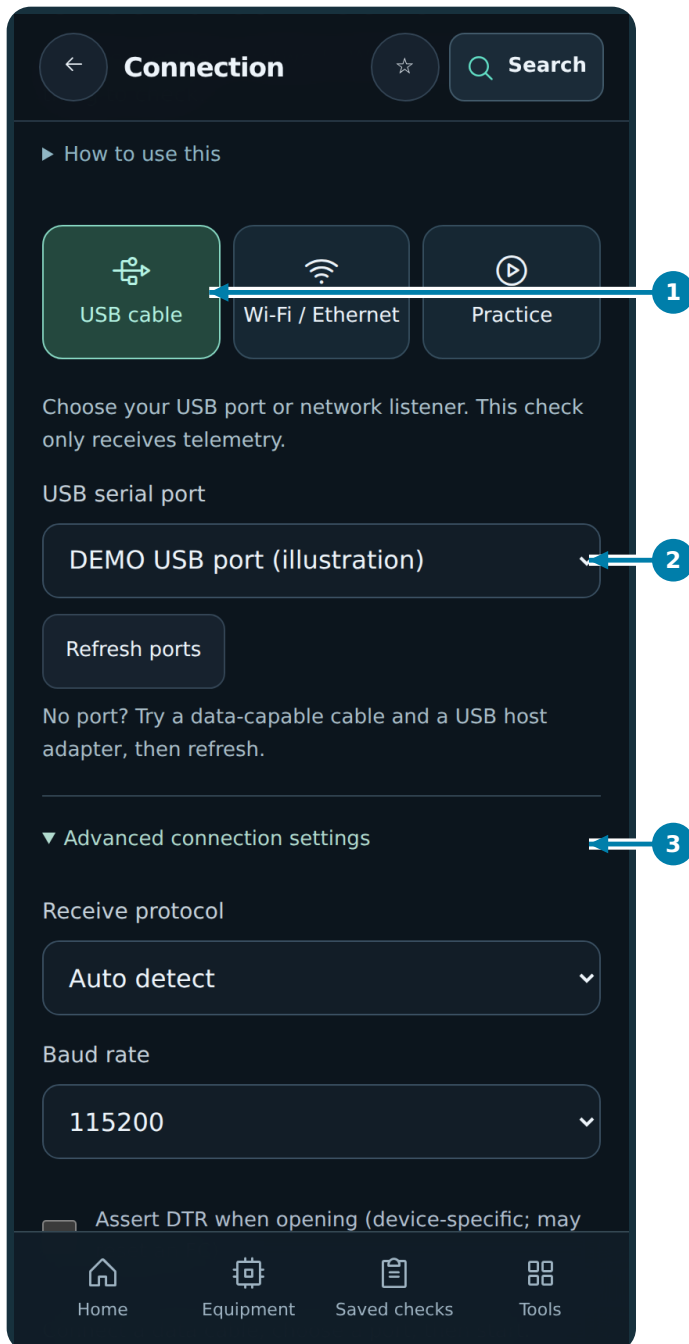
If you changed a shared calculator input, the app marks it inputs changed. Use saved inputs restores the selected equipment's values.

Your saved setups

You can keep up to 50 profiles. They contain the values you entered; selecting one does not detect or configure the hardware.

Check a USB connection

Use this when your phone is connected to a serial telemetry device and you want to see whether data is arriving.



1 Connect, then choose USB cable

Use a cable that carries data and a USB host adapter if your phone needs one. Tap Refresh ports if the device does not appear.

2 Choose the port

Select the device you want to read. Android may ask you to allow access to it.

3 Check the protocol and baud rate

Open Advanced connection settings. Match the device's baud rate; choose Auto detect or the protocol you expect. Scroll down and start the check.

If the connection stays quiet

This checker listens without requesting data. A device using MSP may need requests from another application before it sends anything.

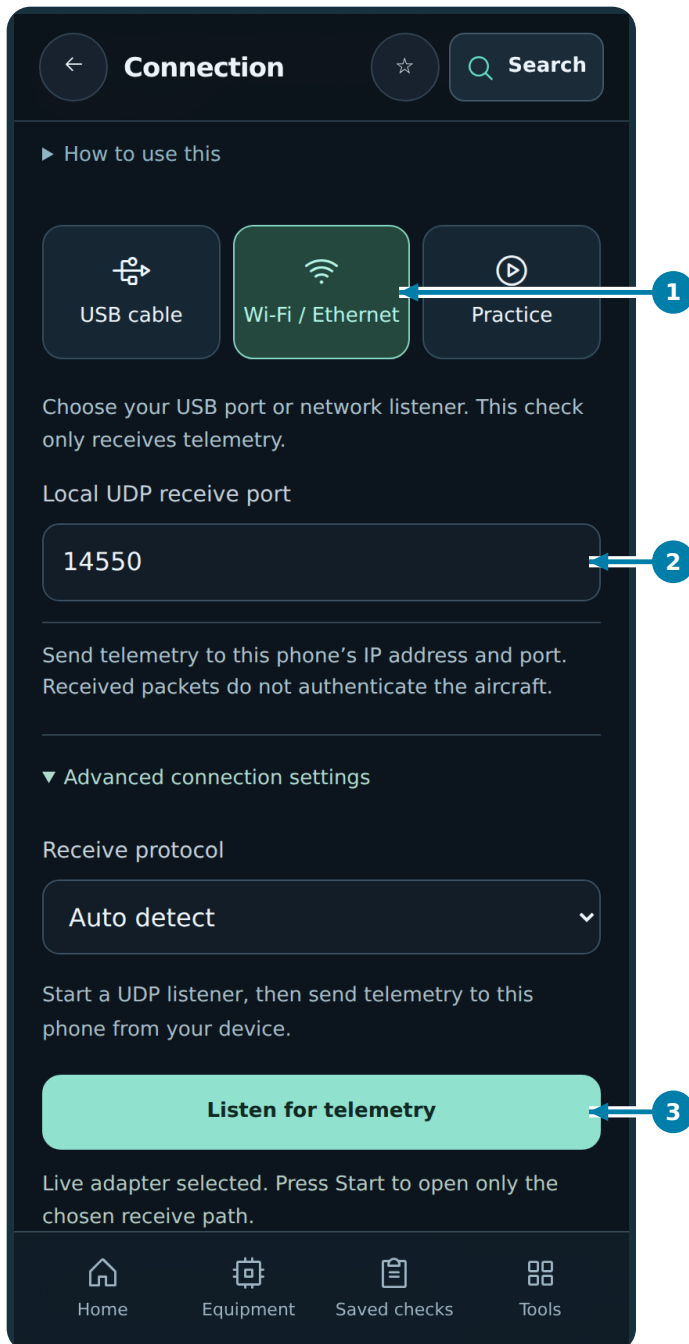
The USB port name in this screenshot is an example.

Before you change adapters

Stop the check first. Pausing the app or changing the adapter or equipment also closes the session. DTR is optional because it can reset some flight controllers.

Listen over Wi-Fi or Ethernet

Use this for a device that sends UDP telemetry to your phone. The phone and sender need a network path to each other.



1 Choose Wi-Fi / Ethernet

This makes the phone listen for incoming packets. It does not find or connect to a flight controller automatically.

2 Match the port at both ends

The default is 14550. On the sending device, set the destination to your phone's IP address and this port.

3 Tap Listen for telemetry

Read the result after data arrives. Tap Stop check when you are finished. Use Practice if you want an example without a sender.

If packets arrive but no position appears

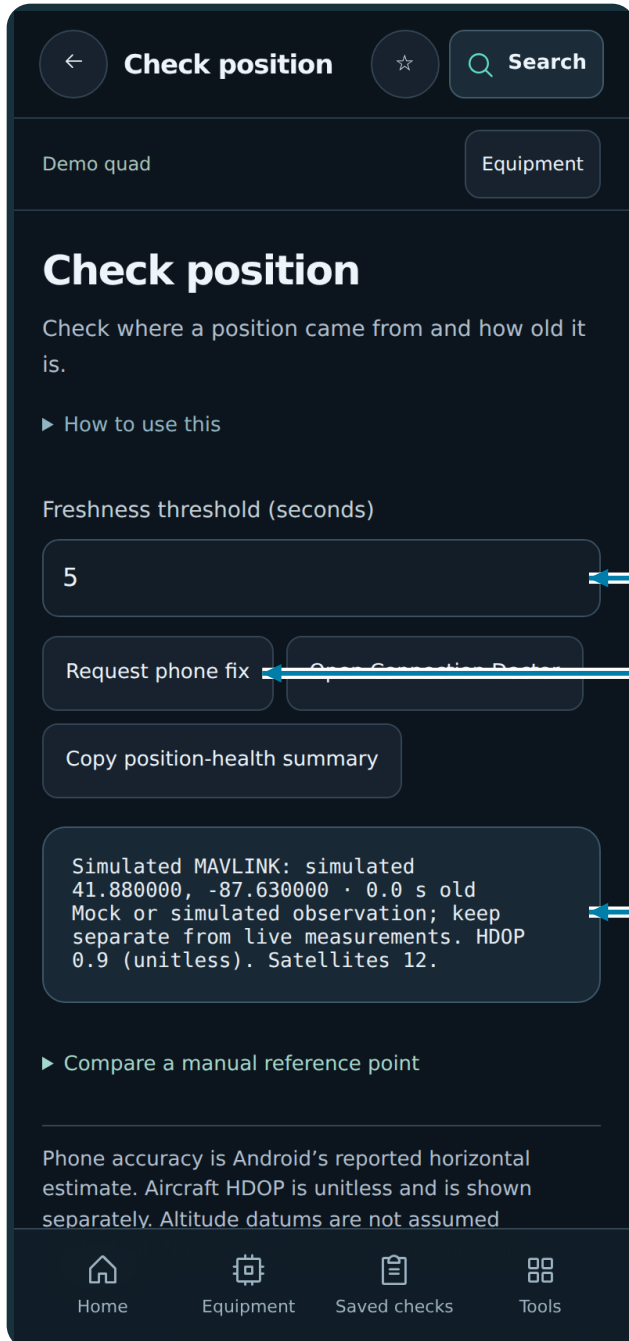
Open the Protocol and Telemetry rows. The sender may be reachable without sending the kind of position data the app understands.

A connection is only part of the check

UDP reception does not verify which aircraft sent it. A TCP video-endpoint check only shows that the port accepts a connection; it does not play video or measure its delay.

Is this position recent enough?

Use this when coordinates are missing, look wrong or may be left over from an earlier connection.



1 Set how old is too old

Enter the maximum age in seconds. Choose a limit that makes sense for what you are checking.

2 Read the source and age

The position may come from the aircraft or a practice session. Check the timestamp and fix information before using it.

3 Compare with the phone

Request phone fix asks Android for a current location. Allow location access if prompted, then try again.

Comparing two positions

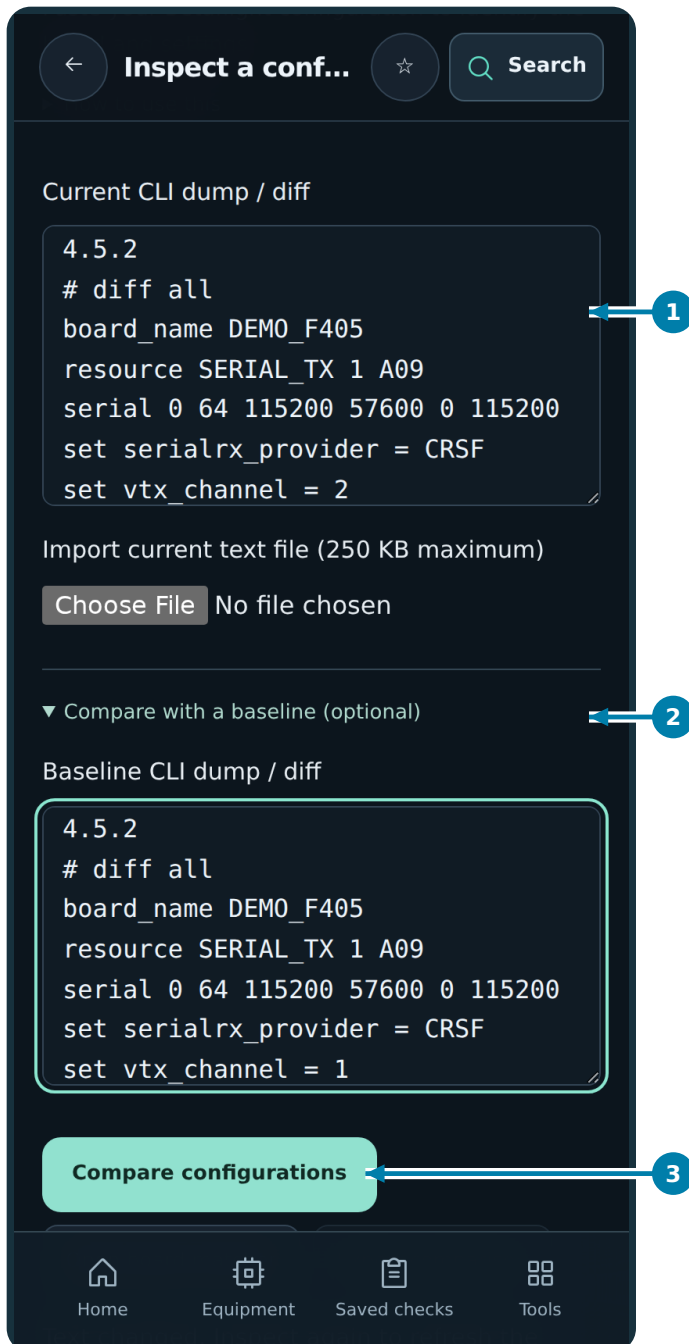
The tool shows phone and aircraft positions separately. You can also enter a known WGS84 position to compare against them.

What a disagreement means

The phone and drone may simply be in different places. HDOP is not an accuracy measurement in meters, and a good GPS fix alone does not establish survey accuracy.

Make sense of a Betaflight dump

Use this when someone sends you CLI text, or when you want to check what changed between two configurations.



1 Paste the current configuration

Use a Betaflight dump or diff, or import a text file. The text stays in this open session.

2 Add an older copy to compare

Open Compare with a baseline and paste the earlier configuration. Prefer a copy from the same board and firmware.

3 Tap Inspect or Compare

The button changes to Compare configurations when a baseline is present.

Start with the board and UARTs

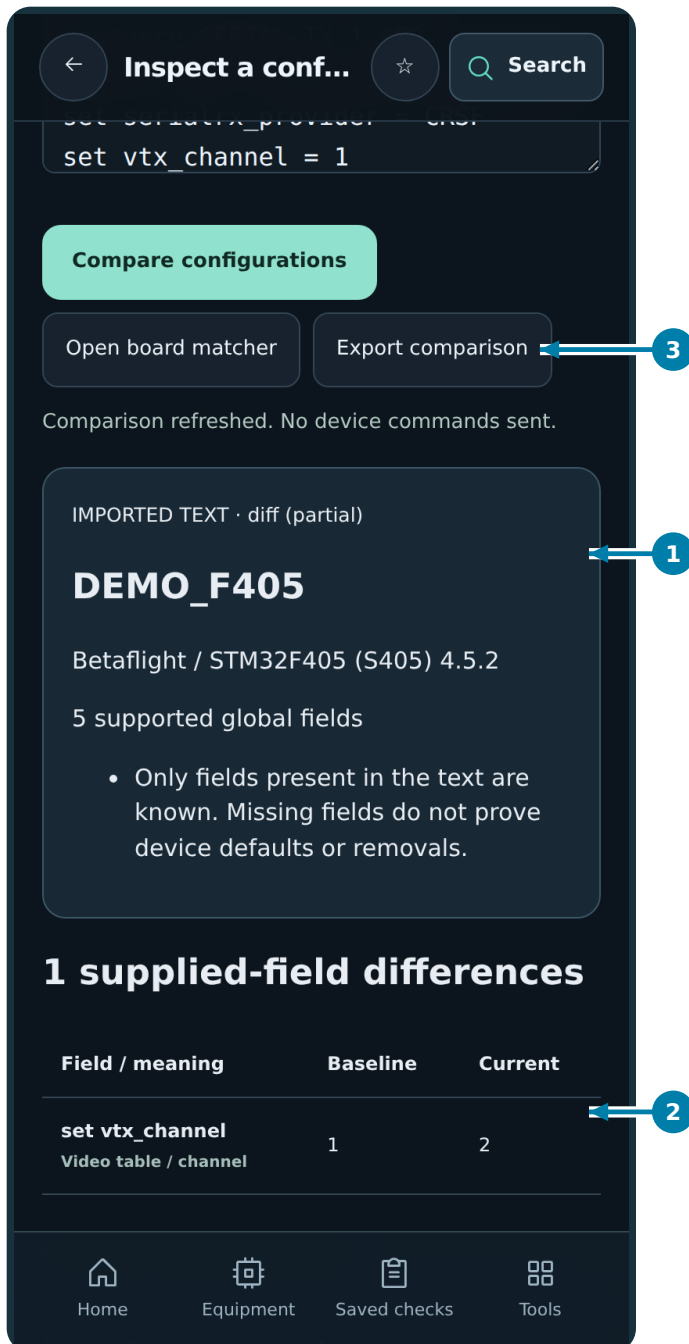
The result pulls out board and firmware names, resource pins, UART settings and VTX-table information. It also flags issues in the supplied text.

A diff can leave things out

A missing line does not mean a setting was removed or reset. Unsupported and profile-specific lines are not fully interpreted. This tool does not send commands to the board.

Find what changed

Read the comparison before assuming that a new configuration caused the problem.



Example board name: DEMO_F405.

Before changing the board

This report is a comparison, not a script to apply. Its SHA-256 hashes identify the supplied text files, not the physical flight controller.

1 Check the board and firmware first

If they differ, the old configuration may not be a suitable comparison. Read any warnings about repeated settings or pin conflicts.

2 Look at the changed fields

Each row puts the old and new values side by side. In this example, the VTX channel changed from 1 to 2.

3 Export if you need a second opinion

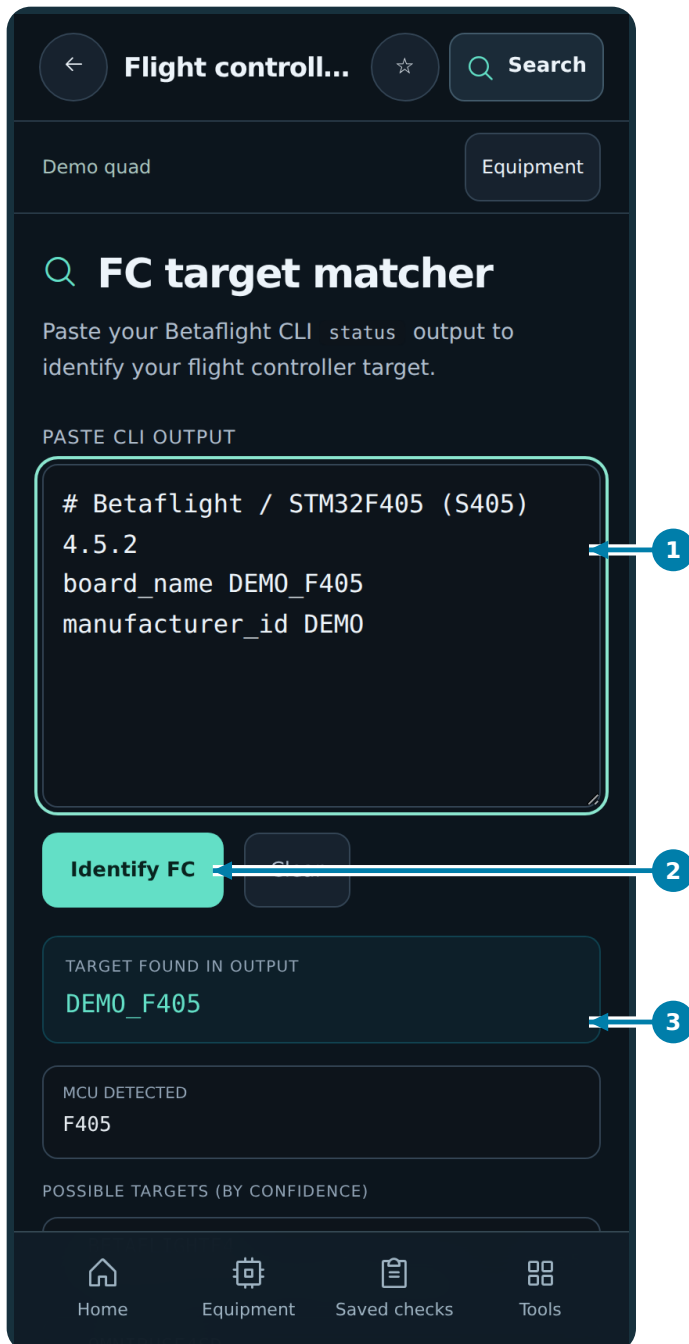
Export comparison saves an HTML report. Send it with a description of the problem you are seeing.

A useful question to ask

Does the changed setting explain the symptom? Open Inspect current global fields when you need to look beyond the differences.

Which flight controller target is this?

Use this when you have Betaflight CLI output but do not know which board documentation to look for.



1 Paste the board's CLI output

Use status or dump-style output from the device you are checking.

2 Tap Identify FC

The matcher looks for the board, manufacturer and other identifying text. Clear starts over.

3 Use the name to find the right documentation

Check that the reported target matches the actual board before choosing firmware or a wiring diagram.

If it cannot identify the board

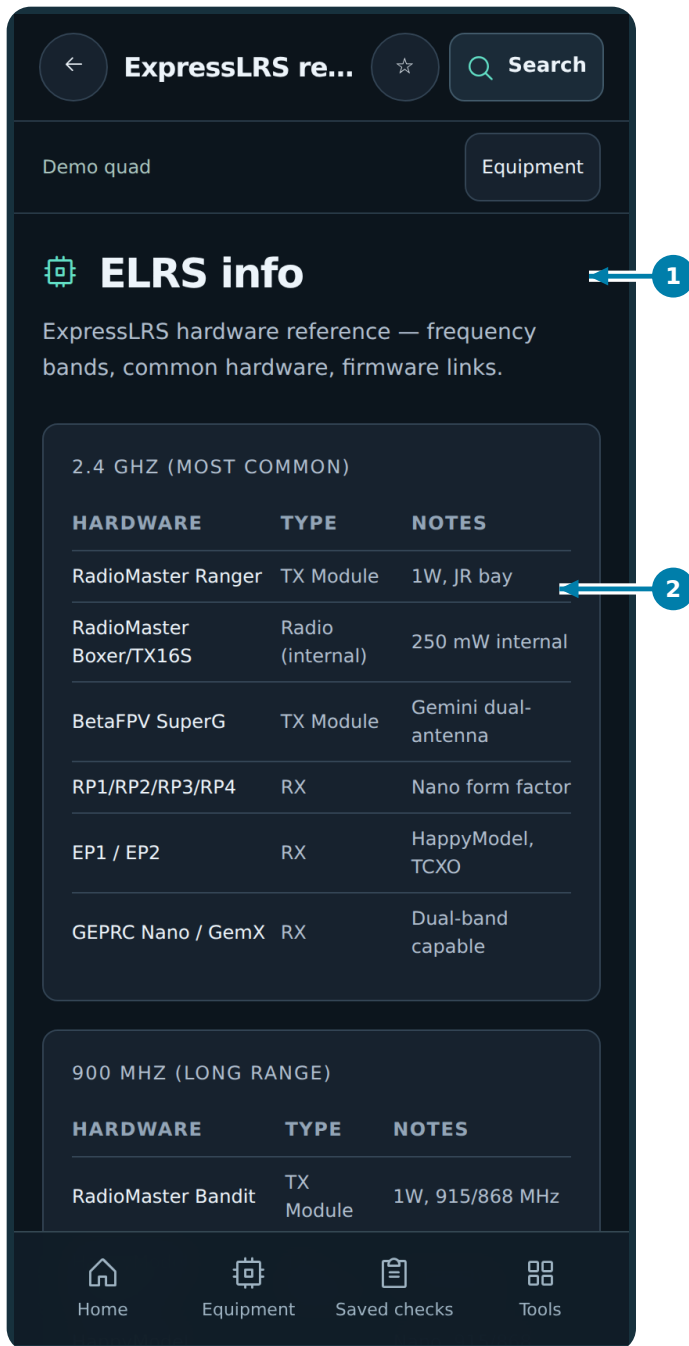
Keep the original output and look for the manufacturer's exact model and revision. An unknown result is better than guessing a similar target.

About this example

DEMO_F405 is a made-up board name used in the guide. The matcher reads text; it does not connect to the board or verify its pin mapping.

Look up an ExpressLRS setting

Use this as a quick reference when you are checking an ELRS radio setup.



1 Open ExpressLRS reference

Find it under Tools. You can read the bundled reference without connecting a radio.

2 Find the section you need

Start with the hardware and band tables. Scroll for packet rates, switch modes and telemetry ratios.

When you need more detail

The page links to the configurator, releases, product finder and signal-health documentation. Those links need internet access.

Check the firmware version

The built-in reference is a snapshot. Confirm current instructions for your exact radio and firmware before changing its setup.

Get a first estimate of radio range

Use this to compare radio setups before you look at terrain or test the link outside.

The screenshot shows the 'Estimate radio range' tool interface. At the top, there's a header with a back arrow, the title 'Estimate radi...', a star icon, and a search bar. Below the header, there's a 'Demo quad' label and an 'Equipment' button. The main section is titled 'Estimate radio range' with a subtitle 'Start with frequency and power. Review radio details for the range estimate.' and a link 'How to use this'. There are two input fields: 'TX power' with the value '100' and 'Frequency' with the value '915'. Below these fields, there's a note: 'The range estimate depends on the radio details below. It does not measure your actual link.' There's a section 'Radio details: gains, sensitivity & margin' with four radio options: 'ELRS 900 100mW', 'ELRS 2.4G 250mW', '5.8G VTX 600mW', and '1.3G VTX 1W'. At the bottom, there's a summary box showing '52.1 km FREE-SPACE LIMIT', '20.0 dBm TX POWER', '22.0 dBm', and '126.0 dB'. The bottom navigation bar has icons for Home, Equipment, Saved checks, and Tools. Numbered callouts 1, 2, and 3 point to the frequency input, the TX power input, and the radio details section respectively.

1 Enter the frequency

Use MHz. For example, enter 915 for a 915 MHz link.

2 Enter transmitter power

This field uses mW. Enter RF output power, not the electrical power the radio draws from its battery.

3 Check the radio details

Expand the gains, receiver sensitivity and margin. Use sensitivity for the radio mode you actually intend to run.

Read it as a starting estimate

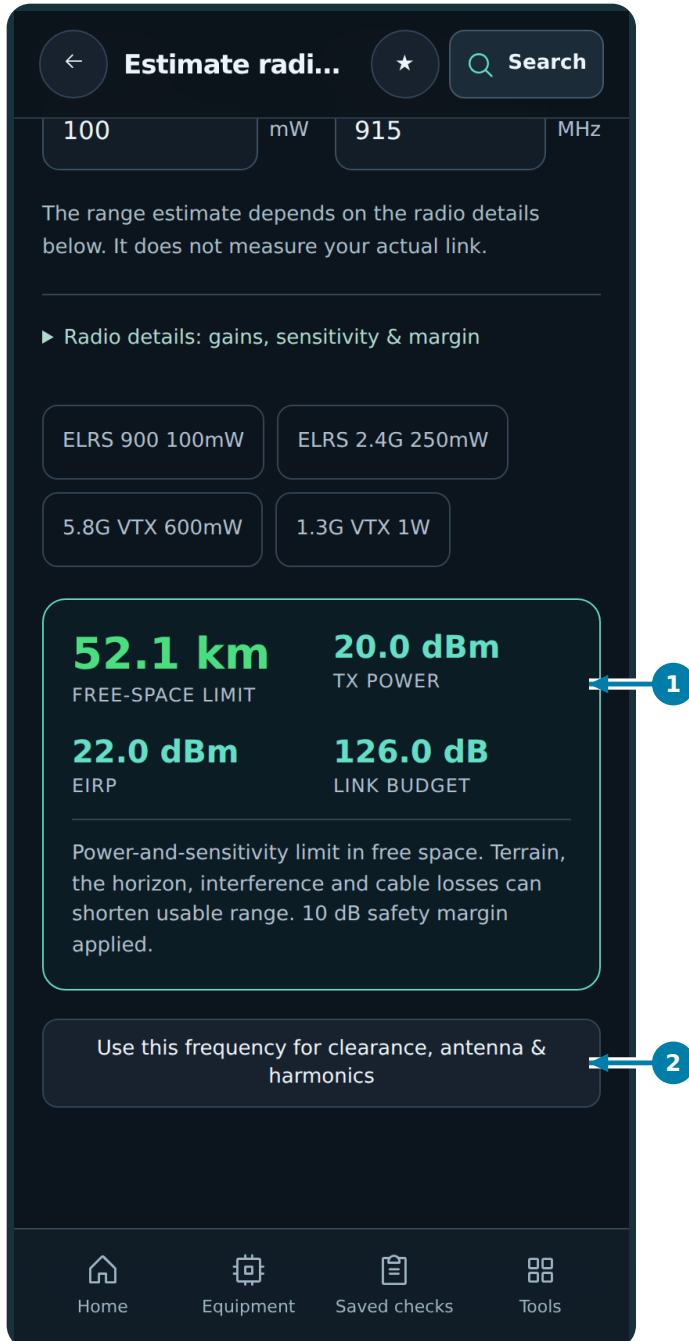
The result includes free-space range and the radio link budget. Scroll down to reuse the frequency in related tools.

Why the real range can be shorter

Terrain, interference, cable loss, antenna placement and the horizon can all reduce the usable range. This calculation has not measured those conditions.

Take the estimate to the next tool

Once the radio numbers look right, check whether the planned path has enough clearance.



1 Read the free-space limit

This is the distance the entered power and sensitivity predict after your selected margin. It is not a tested coverage radius.

2 Tap Use this frequency

This carries the frequency into the Fresnel, antenna-length and harmonics tools so you do not have to enter it again.

Where to go next

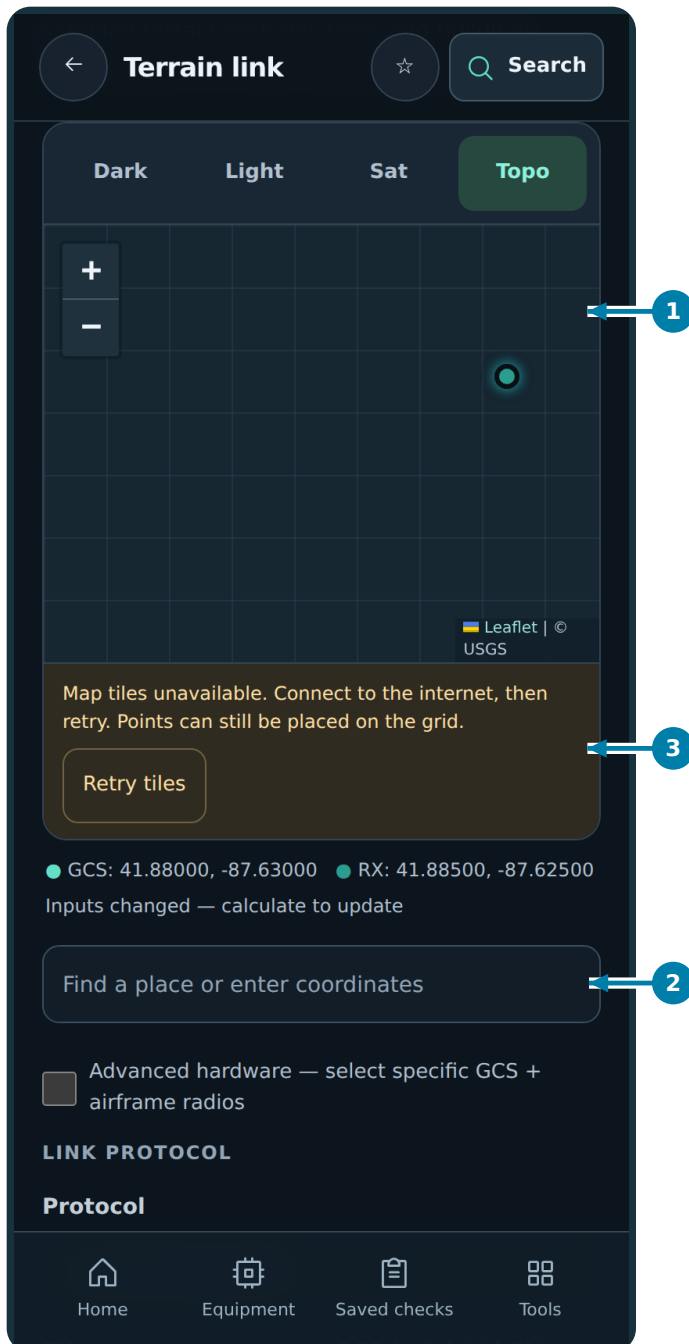
Use Terrain link to look at the route, and Fresnel clearance to see how much space the signal needs around it.

Watch the power units

Link range uses mW. Terrain and mesh use dBm. 100 mW = 20 dBm; 1000 mW = 30 dBm.

Put the radio path on a map

Use Terrain link when a hill or the route itself may be limiting a ground-to-air connection.



1 Place the two ends

Tap once for the ground station and again for the aircraft or receiver. Drag either marker to adjust it.

2 Find the area you need

Enter coordinates or search for a place. The coordinate tool can also center this map.

3 A blank basemap is still usable

You can place points on the grid when map imagery is unavailable. The imagery needs internet access.

Then scroll to the radio settings

Check the selected radios, power, gains, heights and margin before calculating the link.

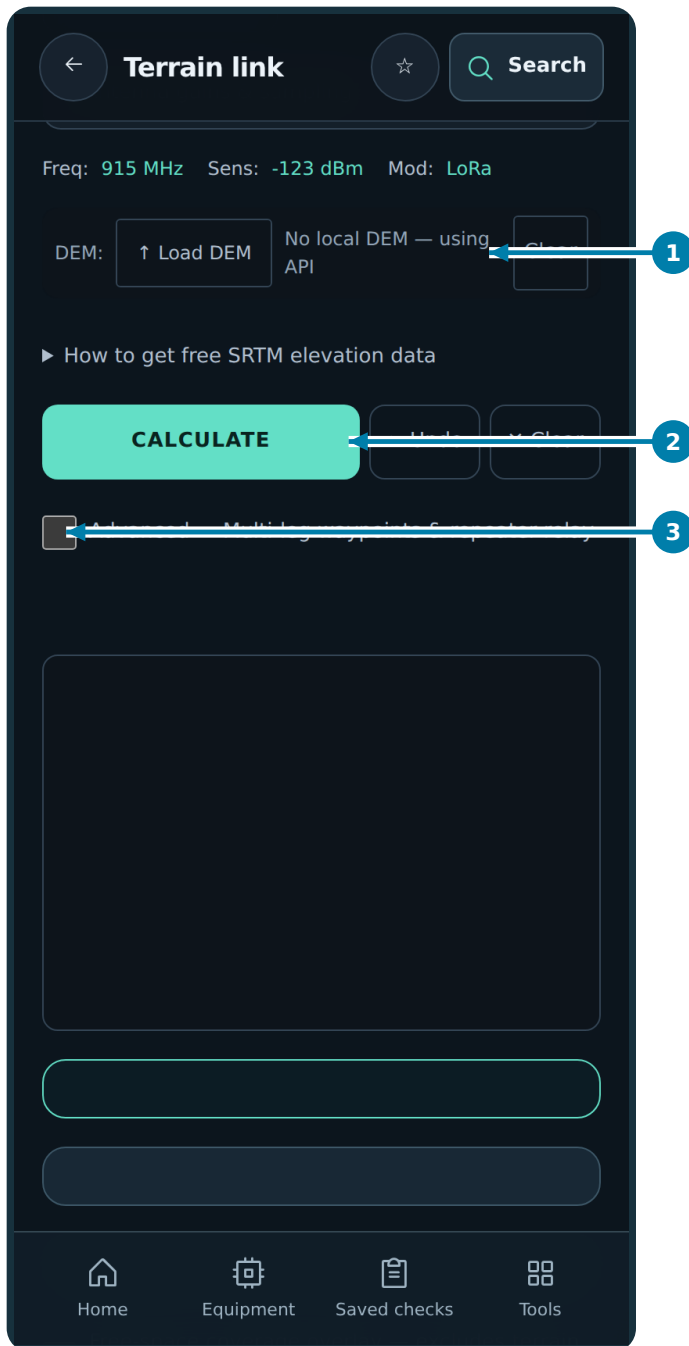
Offline map view with example points.

Map imagery is not elevation data

A blank map does not mean flat ground. Load elevation that covers the route. Equipment profiles do not replace this tool's radio selections.

Check the terrain along the route

With the endpoints placed, give the tool elevation data and review the radio settings.



1 Choose elevation data

Load an SRTM .hgt file or a WGS84 GeoTIFF .tif/.tiff covering the route, or use the available online elevation source.

2 Calculate the link

Check power in dBm, antenna gains and heights. Then read the profile and any problem locations.

3 Try a different route

Advanced mode lets you add waypoints or a repeater path. Compare the legs and export a report when you want to keep the plan.

Look for the part of the route that matters

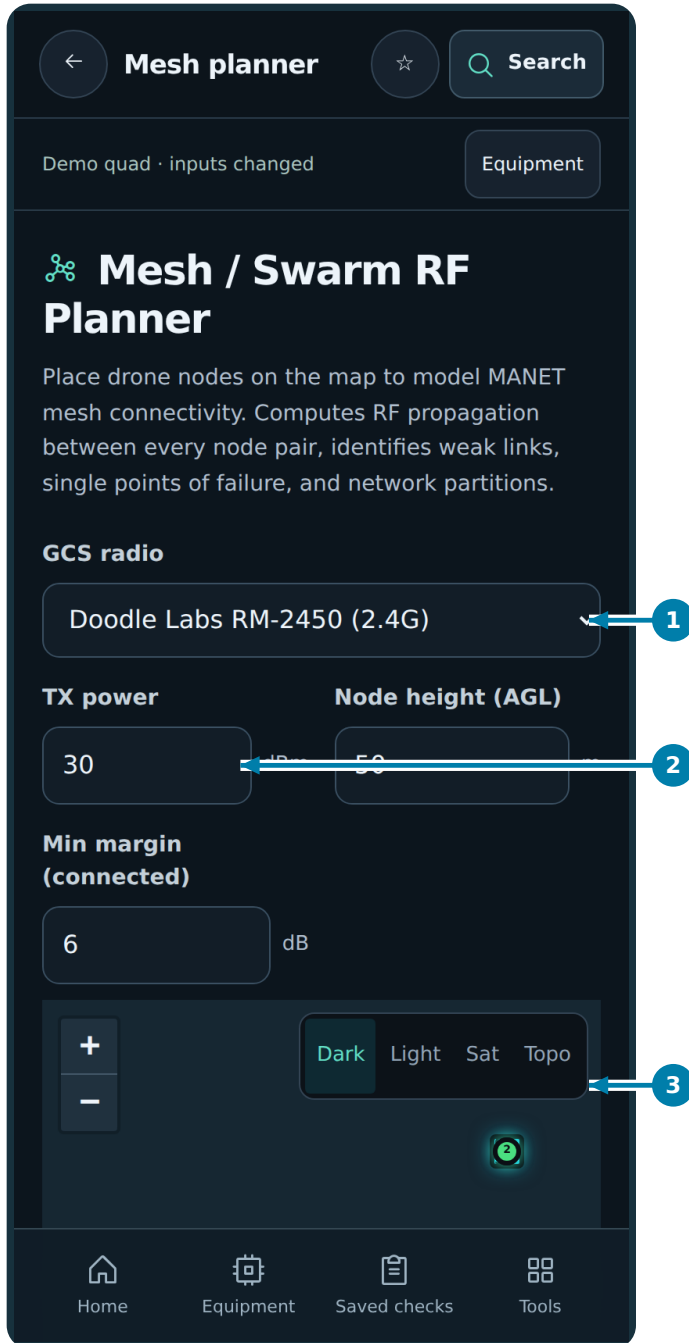
A profile helps you see where the path loses clearance, rather than treating the whole route as one number.

If the data does not cover the path

Load the right area and coordinate system. Missing elevation stays unknown. Even a clear modeled path still needs a real link test.

Sketch a radio network

Use this to compare node locations and see where a planned network may be fragile.



1 Choose the ground-station radio

Pick the intended radio family or Custom. You can adjust individual node radios below the map.

2 Set power, height and margin

Power is in dBm. Use realistic heights and a margin that reflects the link you want to plan.

3 Add the nodes

Tap for the ground station first, then the other nodes. Drag markers to try another layout.

Plan before placing hardware

The tool lets you compare links and groups of connected nodes before you build the network.

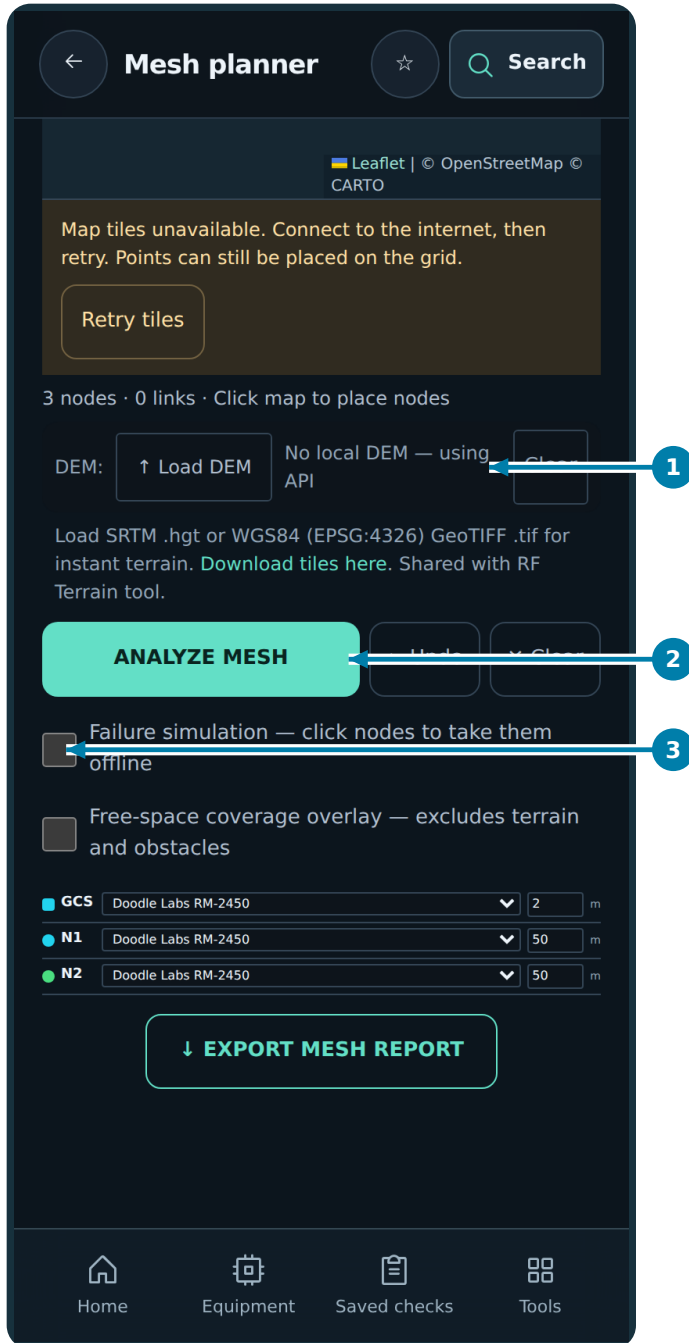
Example node layout on the offline grid.

Use compatible radios

Choosing presets does not make different protocols work together. This map is your plan; it has not discovered nearby radios.

What if one node goes down?

A network that looks connected can still depend on a single relay. Try losing that relay in the model.



1 Load terrain for the area

Mesh and Terrain link share the elevation controls. Missing data stays unknown.

2 Tap Analyze mesh

Review the link margins and connectivity table. Open a link for more detail or export the report.

3 Try Failure simulation

Enable it, then tap a node to take it offline in the plan. Look for nodes that become disconnected.

Look for a weak point

If losing one relay separates the network, try another location or another path. Compare the layouts before testing them outside.

Read the model for what it is

Coverage overlays exclude terrain and obstacles. Hop counts and latency are estimates, not measurements from a running network.

How much clearance does the link need?

Radio signals need space around the straight line between antennas. This tool gives you a scale for that space.

← Check path cl... ☆ Search

Demo quad Equipment

Check path clearance

Check the clear space needed around a radio path.

► How to use this

Link distance **Frequency**

1000 015 MHz

Point along path

50 %

9.1 m
ZONE 1 RADIUS AT
50%

5.4 m
60% CLEARANCE
NEEDED

32.8 cm
WAVELENGTH

TX RX

Home Equipment Saved checks Tools

1 Enter the full path length

Use meters for the distance between the two ends.

2 Enter frequency and position

Frequency is in MHz. Position along path chooses where you want to calculate the radius.

3 Read the two clearance numbers

The full first-zone radius and the suggested 60% clearance are different. Use the diagram to keep them straight.

Use it alongside the terrain profile

Compare the clearance with antenna heights, nearby trees and the ground along the path.

You still need to know what is there

This calculator does not locate trees, buildings or other obstacles. It tells you how much space to look for.

Work out an antenna starting length

Use this when planning a dipole, or comparing how antenna dimensions change with frequency.

The screenshot shows the 'Antenna length' tool interface. At the top, there's a navigation bar with a back arrow, the title 'Antenna length', a star icon, and a search bar. Below this, there's a 'Demo quad' label and an 'Equipment' button. The main section is titled 'Antenna length' with a subtitle: 'Calculate quarter-wave and half-wave antenna element lengths for any frequency. Account for velocity factor of coax or wire.' Below the subtitle is a link '► How to use this'. The input section has two fields: 'Frequency' (set to 915 MHz) and 'Velocity factor' (set to 0.67). Below these is a 'Unit' dropdown menu set to 'mm'. There are also several preset frequency buttons: 433 MHz, 868 MHz, 915 MHz, 1.3 GHz, 2.4 GHz, and 5.8 GHz. The output section shows four results: '79.5 mm QUARTER-WAVE ($\lambda/4$)', '159.0 mm HALF-WAVE ($\lambda/2$)', '198.8 mm', and '318.0 mm'. At the bottom is a navigation bar with icons for Home, Equipment, Saved checks, and Tools. Numbered callouts point to specific elements: 1 points to the Frequency input field, 2 points to the Velocity factor input field, and 3 points to the output results section.

1 Set the frequency

Enter the intended band in MHz. Use this frequency in the range tool can fill it for you.

2 Check velocity factor and units

Use a factor suitable for the antenna material and construction. Choose the units you want to work in.

3 Read the element lengths

The tool shows wavelength and the calculated lengths for your selected inputs.

A length to start from

Use the dimensions as a starting point for construction and tuning.

Tune the finished antenna

The feed and installation affect the final result. This tool does not measure SWR or impedance.

Could these frequencies overlap?

Use this when choosing bands or investigating a possible interference problem.

The screenshot shows the 'Harmonics' tool interface. At the top, there is a back arrow, the title 'Harmonics', a star icon, and a search bar. Below this are three tabs: 'TX frequency', 'Video band', and 'GNSS bands'. The 'TX frequency' tab is active. It contains two input fields: 'TX frequency' with the value '915' and 'TX power' with the value '37'. Below these are two more input fields: 'Video band start' with the value '5600' and 'Video band end' with the value '5950'. A green checkmark icon and text state: '✓ No nearby GNSS harmonics found in this frequency check at 915 MHz'. Below this is a green box with a checkmark icon and text: 'No video band conflicts'. Below that is a section titled 'VIDEO BAND CONFLICTS (5600-5950 MHZ)' with the text 'No harmonics in video band at this frequency.' At the bottom of the screen is a navigation bar with four icons: 'Home', 'Equipment', 'Saved checks', and 'Tools'. Three numbered callouts are present: 1 points to the 'TX frequency' input field, 2 points to the 'Video band start' input field, and 3 points to the 'No video band conflicts' green box.

1 Enter the transmitter frequency

Use MHz for frequency and dBm for power, or choose a suitable preset.

2 Set the video band

Enter its start and end frequencies. The end must be higher than the start.

3 Read the flagged multiples

Scroll through the harmonics, video-band overlaps and GNSS warnings.

A place to start investigating

A flagged harmonic tells you which frequency relationship deserves a closer look.

Confirm interference with measurements

Suppression and risk levels are approximate. An overlap in the calculation does not prove that the transmitter is causing interference.

Prepare settings for a video transmitter

Use this to generate Betaflight CLI text for a supported VTX model.

The screenshot shows the 'VTX configura...' screen in the Prismo Field Kit. At the top, there's a back arrow, a star icon, and a search bar. Below this, 'Demo quad' is displayed next to an 'Equipment' button. The main section is titled 'VTX config generator' with a subtext: 'Generate ready-to-paste Betaflight vtxtable CLI commands for your VTX. Select your model, copy, paste into CLI.' There are three main selection areas: 'Manufacturer' (set to 'Foxeer'), 'Model' (set to 'Reaper Extreme 2.5W 40Ch'), and 'UART port' (set to 'UART 2'). Below these, a note says 'Choose an exact supported model. Tables and power values come from Betaflight presets. [Preset source](#) · [Foxeer Reaper Extreme 2.5W 40Ch](#)'. A section labeled 'GENERATED CLI COMMANDS' contains a 'Copy' button. At the bottom, a text box shows the generated command: '# Foxeer Reaper Extreme 2.5W 40Ch - IRC'. The bottom navigation bar has icons for Home, Equipment, Saved checks, and Tools. Three numbered callouts are present: 1 points to the 'Model' dropdown, 2 points to the 'UART port' dropdown, and 3 points to the 'Copy' button.

1 Choose the manufacturer and model

Match the exact device. The source link shows where the preset settings came from.

2 Choose the UART

Check which UART the VTX control wire actually uses and which protocol the hardware supports.

3 Copy the text

Review the generated settings, then tap Copy. You can use the text in your usual Betaflight configuration workflow.

A head start on the configuration

The tool prepares settings and a VTX-table snippet. You still decide whether they are right for the board.

Before you paste it into Betaflight

Check the board, firmware, wiring and supported frequency and power settings. Field Kit does not apply these commands.

Build a Betaflight VTX table

The app calls this Unlock VTX table. It creates the channel and power table text for you to review.

The screenshot shows the 'VTX table' app interface. At the top, there's a back arrow, the title 'VTX table', a star icon, and a search bar. Below this is a 'CONFIGURATION' section. The first option is 'VTX Protocol' with a dropdown menu currently showing 'SmartAudio 2.1 (dBm)'. The second option is 'Max Power Level' with a dropdown menu showing '25 / 200 / 500 mW'. The third option is 'Bands to include', which has a list of checkboxes: 'Raceband (R)' (checked), 'Fatshark (F)' (checked), 'Boscam E' (checked), 'Boscam A' (unchecked), and 'Boscam B' (unchecked). Below the configuration section is a large green button labeled 'Copy to clipboard'. At the bottom of the screen is a 'BETAFLIGHT CLI OUTPUT' section with a text area showing generated text: '# Generic 5.8 GHz table – verify power levels against your VTX manual' and '# Protocol: SmartAudio 2.1 (dBm)'. The bottom navigation bar has four icons: Home, Equipment, Saved checks, and Tools.

1 Choose the control protocol

2 Choose power options and bands

3 Copy the table

1 Choose the control protocol

Use the protocol your VTX and firmware support. The power-value format changes with the protocol.

2 Choose power options and bands

Select the maximum-power table option and the bands you want included.

3 Copy the table

Read the generated text, then copy it for your Betaflight workflow.

A table you can inspect before using

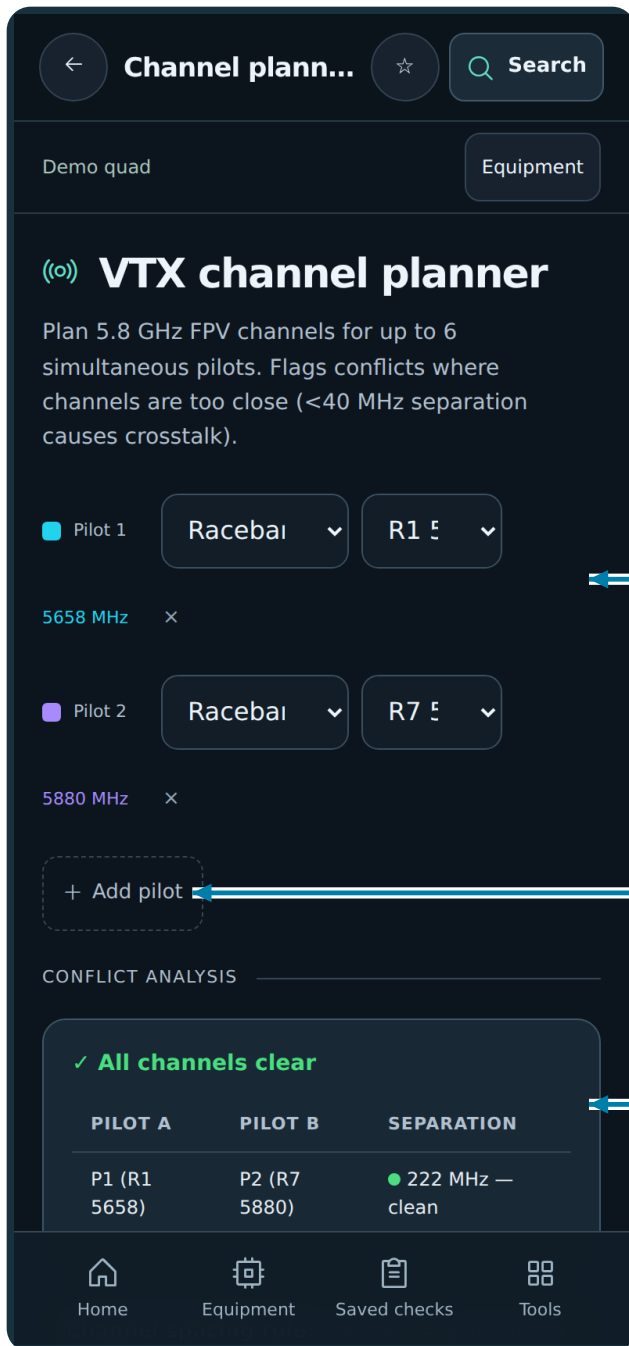
The output lists channel frequencies and power entries for the options you selected.

The name can be misleading

Generating a table does not unlock hardware. Confirm which frequencies and power levels your VTX supports and you can use.

Give each pilot a video channel

Use this when up to six pilots will use video transmitters at the same time.



1 Enter each planned channel

Choose a band and channel for each pilot. Check that transmitter and receiver use the same channel table.

2 Add the rest of the pilots

Tap Add pilot for more rows, up to six. Remove any rows you do not need.

3 Adjust channels that are too close

Read the spacing warnings, change the assignments and check again.

A channel plan to take to the group

Use it to spot close frequencies before everyone switches their video transmitters on.

Test the whole setup together

The app flags spacing below 40 MHz. Real interference also depends on transmitter filtering, power and physical placement.

Find the channel closest to a frequency

Use this when you know a video frequency but need the corresponding band and channel name.

Closest channel Search

Demo quad Equipment

Closest channel finder

Find the nearest named FPV channel to any target frequency. Useful when operating on non-standard or modified bands.

Target frequency

5800

Band filter

Raceband (R)

R5
CLOSEST CHANNEL

5806 MHz
CHANNEL FREQUENCY

±6 MHz
OFFSET

CHANN EL	BAND	FREQ (MHZ)	DISTAN CE
R1	Raceba nd	5658	142 MHz

Home Equipment Saved checks Tools

1 Enter the frequency in MHz

The tool checks the channel tables included in the app.

2 Choose the bands to search

Filter to Raceband or another band, or include all bands if you are unsure.

3 Check the frequency difference

The result shows the closest entry. The table below lets you compare other choices.

Match names at both ends

Use the result to help line up the transmitter and receiver settings.

Closest does not always mean usable

Confirm the actual channel tables on both devices. This tool looks up frequencies; it does not scan for a signal.

Put a location in the format you need

Use this to move between decimal degrees, degrees/minutes/seconds and MGRS.

The screenshot shows the 'Convert coordinates' tool interface. At the top, there's a header with a back arrow, the title 'Convert coord...', a star icon, and a search bar. Below this is a 'Demo quad' label and an 'Equipment' button. The main section is titled 'Convert coordinates' with a subtitle 'Enter a location, or use your phone. Copy it in the format you need.' and a link 'How to use this'. A 'Location' input field contains '48.8582, 2.2945'. Below it is a 'Convert' button. The output section, titled 'WGS84 · Manual entry', displays three formats: '48.858200, 2.294500', '48°51'29.52"N 2°17'40.20"E', and '31UDQ4825111932'. A note at the bottom states 'Grid output has 1 m cells; that is formatting precision, not measured accuracy.' The bottom navigation bar includes icons for Home, Equipment, Saved checks, and Tools. Three numbered callouts are present: 1 points to the input field, 2 points to the 'Convert' button, and 3 points to the output section.

1 Paste the location

Enter decimal degrees, DMS or MGRS. Decimal degrees are latitude first, longitude second.

2 Convert, or use phone location

Manual conversion works offline. Use phone location asks Android for a recent fix.

3 Copy the format you need

Read DD, DMS and MGRS in the result. Scroll down for Copy coordinates or Open in terrain.

Move the location into your plan

Open in terrain centers the map on it. You still place the path endpoints yourself.

More digits do not mean a better fix

A 1 m MGRS cell or six decimal places describes formatting, not measured accuracy. MGRS conversion covers 80°S to 84°N.

How long might the battery last?

Use this for a rough estimate when you know capacity and typical current draw.

The screenshot shows a mobile app interface for estimating battery runtime. At the top, there's a header with a back arrow, the title 'Estimate batt...', a star icon, and a search bar. Below this is a 'Demo quad' label and an 'Equipment' button. The main section is titled 'Estimate battery time' with a subtitle 'Enter capacity and average current to estimate usable time.' and a link 'How to use this'. There are four input fields: 'Rated capacity (mAh)' with a value of 5000, 'Nominal pack voltage (V)' with a value of 22.2, 'Average current (A)' with a value of 20, and 'Reserve (%)' with a value of 20. Below these is a large box for the 'ESTIMATED USABLE RUNTIME' showing '12.0 min'. Underneath the runtime, it says '4000 mAh usable · 20% held in reserve' and '111.0 Wh nominal · 88.8 Wh usable'. At the bottom, a note states 'Based on 20 A average draw. Measure current under the intended operating conditions.' The bottom navigation bar has icons for Home, Equipment, Saved checks, and Tools. Three numbered callouts are present: 1 points to the 'Rated capacity' field, 2 points to the 'Average current' field, and 3 points to the 'ESTIMATED USABLE RUNTIME' result.

1 Enter the battery rating

Capacity is in mAh and nominal voltage is in V. Check the label for the pack you will use.

2 Enter average current

Use amps, not milliamps. A measured average under the intended conditions is more useful than a brief peak.

3 Choose a reserve and read the time

The reserve holds back some capacity. The result also shows energy in watt-hours.

A worked example

5000 mAh at 20 A, with a 20% reserve, gives 12 minutes. At 22.2 V, the pack has 111 Wh of nominal energy.

Use it to plan, then measure

Battery health, voltage sag, temperature and changing loads affect real runtime. This is not a live remaining-flight-time display.

Make sense of a signal reading

Use this with readings from your radio. You can also convert transmitter power between mW and dBm.

The screenshot shows a mobile app interface for checking signal readings. At the top, there's a title 'Check signal r...' with a back arrow, a star icon, and a search bar. Below the title, there are two input fields: 'Measured RSSI (dBm)' with the value '-85' and 'Selected sensitivity (dBm)' with the value '-105'. A green box highlights the 'Selected sensitivity' field, with callout 1 pointing to the field and callout 2 pointing to the up/down arrows. Below these fields is a 'Measured link quality (%)' field with the value '100'. A large box displays the calculated headroom: '100 mW = 20.00 dBm', 'MANUAL READING · RSSI HEADROOM', '20.0 dB', '-85 dBm RSSI - (-105 dBm sensitivity)', and 'Link quality: 100% of packets received'. Callout 3 points to this large box. At the bottom, there's a footer with icons for Home, Equipment, Saved checks, and Tools. A link 'ExpressLRS signal-health reference' is also present.

1 Enter RSSI in dBm

Use a reading from the radio you are checking.

2 Enter the right receiver sensitivity

Use the value for that radio's mode and packet rate. Changing the mode can change the sensitivity.

3 Read headroom and link quality separately

Headroom is RSSI minus sensitivity. Link quality describes how many packets were received.

A worked example

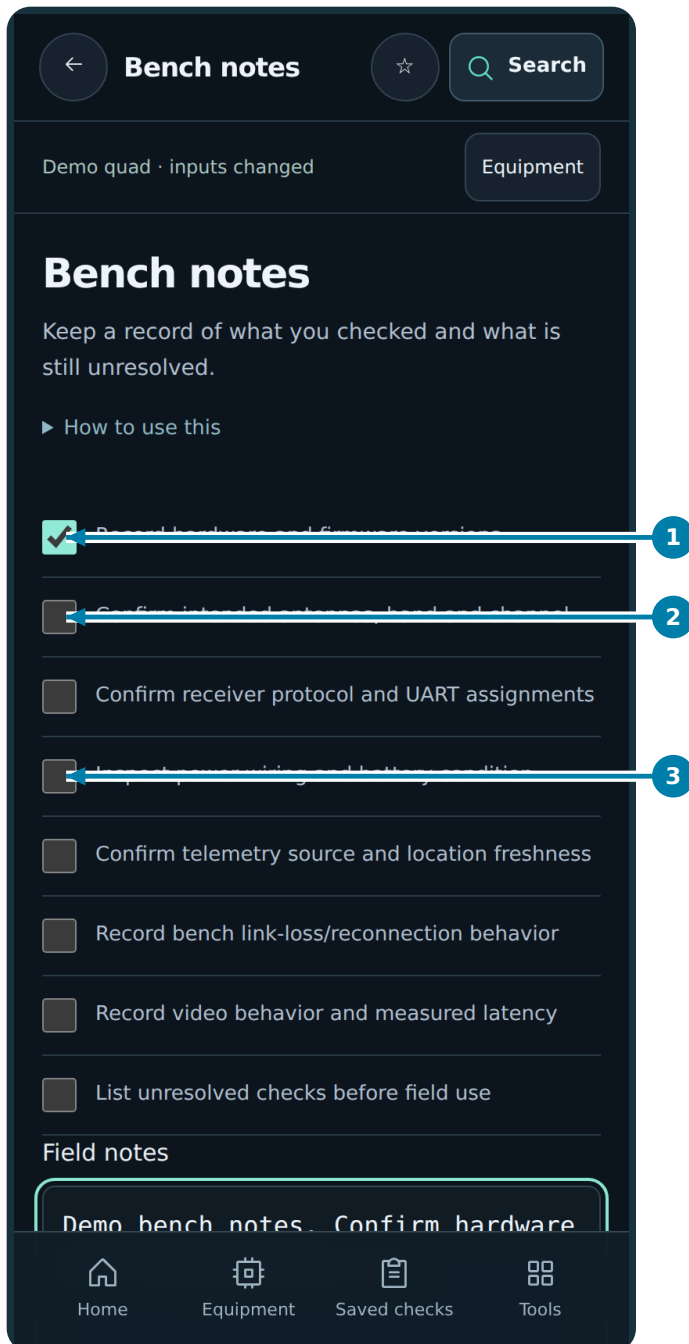
RSSI of -85 dBm and sensitivity of -105 dBm gives 20 dB of headroom. You can convert mW and dBm in either direction above.

These are your entered readings

The app does not measure RF here. A good headroom number alone does not guarantee a reliable link or a particular range.

Keep track of what you checked

Use this at the bench so the next person, or your future self, can see what is finished and what is still open.



1 Start with hardware and firmware

Record the versions before you troubleshoot. They help you compare with earlier work.

2 Check the intended radio setup

Confirm antennas, band and channel. Make clear whether a setting is only planned or has been checked on the equipment.

3 Work through the rest

Scroll for UARTs, power wiring, telemetry age, reconnection, video and unresolved questions.

A checklist you can come back to

The app keeps your ticks and notes on this device. The count shows how many items you marked.

Two kinds of records

You tick these boxes yourself. They are separate from the snapshots saved by the connection checker, and the app does not verify an item just because it is ticked.

Leave useful bench notes

Scroll below the checklist. A few specifics are much more helpful than “tested” or “does not work.”

Bench notes Search

- ☐ Confirm receiver protocol and UART assignments
- ☐ Inspect power wiring and battery condition
- ☐ Confirm telemetry source and location freshness
- ☐ Record bench link-loss/reconnection behavior
- ☐ Record video behavior and measured latency
- ☐ List unresolved checks before field use

Field notes

Demo bench notes. Confirm hardware and firmware before use.

1 of 8 recorded · self-reported, not certification

Export checklist Start new checklist

Home Equipment Saved checks Tools

1 Write what you actually tried

Name the equipment and firmware, describe the connections and include any measurements or open questions.

2 Check the completed count

The count reflects the boxes you ticked. Leave an item unticked if you have not checked it.

3 Export before starting over

Export checklist saves an HTML file. Start a new checklist clears the current ticks and notes after confirmation.

A note worth keeping

For example: “USB data arrived at 115200 baud. GPS age stayed above 30 seconds. Video was checked separately; latency was not measured.”

Keep the file if you need it later

Clearing app storage removes local checklist notes. Export first, then archive or share the file.

Find an earlier check

Use Saved checks before repeating work. An earlier result may explain when a problem started or what someone already tried.

1 Match the name and serial number

Each entry keeps the equipment settings used when that check was saved.

2 Read the time and type of check

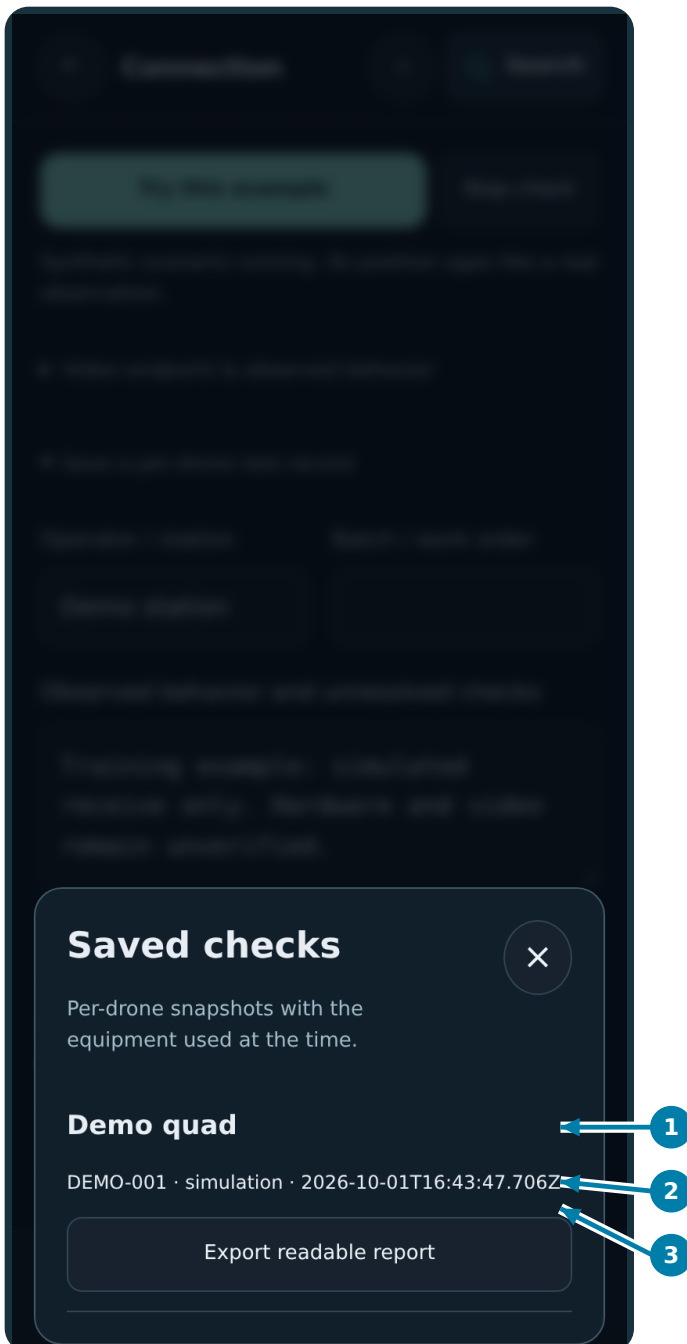
Practice stays labeled as simulation. Older records do not change when you edit the current equipment setup.

3 Export the one you need

Export readable report creates an HTML file with the results and notes. Extra details expand farther down.

Compare with what you see today

Look for changes in firmware, cabling, radio mode or test conditions. Configuration comparisons and bench checklists have their own exports.



What stays on this phone

The app keeps the last 50 snapshots. File hashes help identify the saved data; they do not verify an aircraft's identity or certify a production test.

What the other person receives

This is a real HTML export from the practice check shown earlier. It opens in a browser.

Equipment diagnostic record

2026-10-01T16:43:47.706Z

Demo quad · DEMO-001

Equipment revision 1 · simulation

Diagnostic observations only. Simulations and manual notes are labeled; unverified steps remain open.

Check	Status	Observation
Transport	observed	Software fixture · healthy
Protocol	observed	MAVLINK · 180 bytes · 3 counted MAV/GHST/CRSF frames
Telemetry	simulated	Mock or simulated observation; keep separate from live measurements. Age 1.1 s.
Video	unverified	Simulated stream available; no device video or latency

1 Check the name and type of check

The report names Demo quad and serial DEMO-001. It also says simulation, so the example cannot be mistaken for a hardware check.

2 Read the results and notes

Each row shows what was checked, its status and what the app saw. Scroll for the rest of the rows, notes and full record.

No app needed to read it

Someone can open this file without installing Field Kit. Send the HTML file itself, rather than only a screenshot of the report.

HTML report from the example practice session.

Keep the context with the result

This report came from Practice. For your own checks, include the connections and test conditions in the notes.

When something is not working

Start with the connection and earlier notes. You do not need to change everything at once.

Before repeating an old problem

Look through Saved checks and earlier exports for the same serial number. See what was tried, what firmware was used and what changed. It may save you a second round of the same troubleshooting.

The USB device does not appear

Check that the cable carries data, the phone supports the connection and Android granted USB access. Tap Refresh ports. If a port appears but no data arrives, check protocol, baud rate and whether the device actually sends data without requests.

Network data or GPS is missing

Check the sender's destination IP and port. Open the protocol details if bytes arrive but cannot be decoded. Fresh packets do not guarantee fresh GPS: check the position timestamp too.

The map is blank

You can still place points on the grid. Map pictures and elevation are separate: load a supported elevation file covering the route for offline terrain calculations.

Before clearing app storage

Export the records you need. Equipment, calculator inputs, pinned and recent tools, checklist notes and the latest 50 snapshots are stored on this device. Raw configuration text lasts only for the open session.

About the screenshots

These screens come from Field Kit 0.3.1, package com.dronewukong.fieldtools. Demo quad, DEMO-001, DEMO_F405 and the USB port label are examples. Practice uses software-generated data. This guide does not claim a physical hardware test.

Keep the right record

Connection snapshots, configuration comparisons, bench checklists, terrain plans and mesh plans each have their own export. Keep the one that answers the question you are working on.